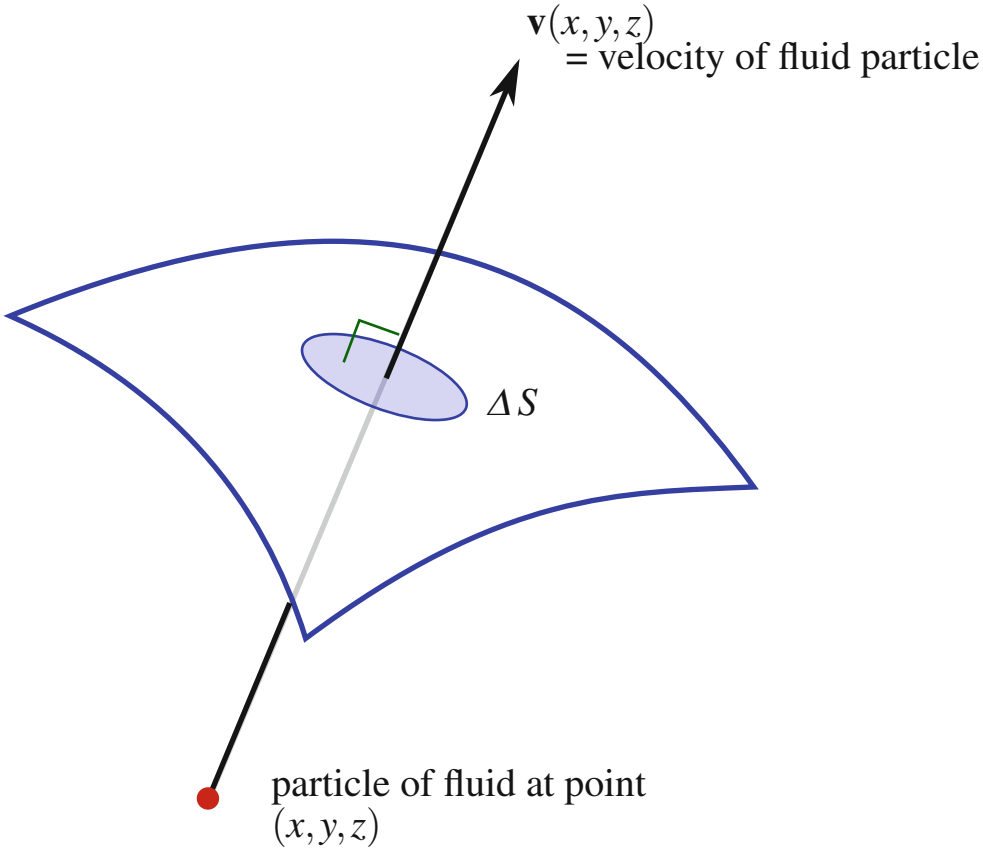
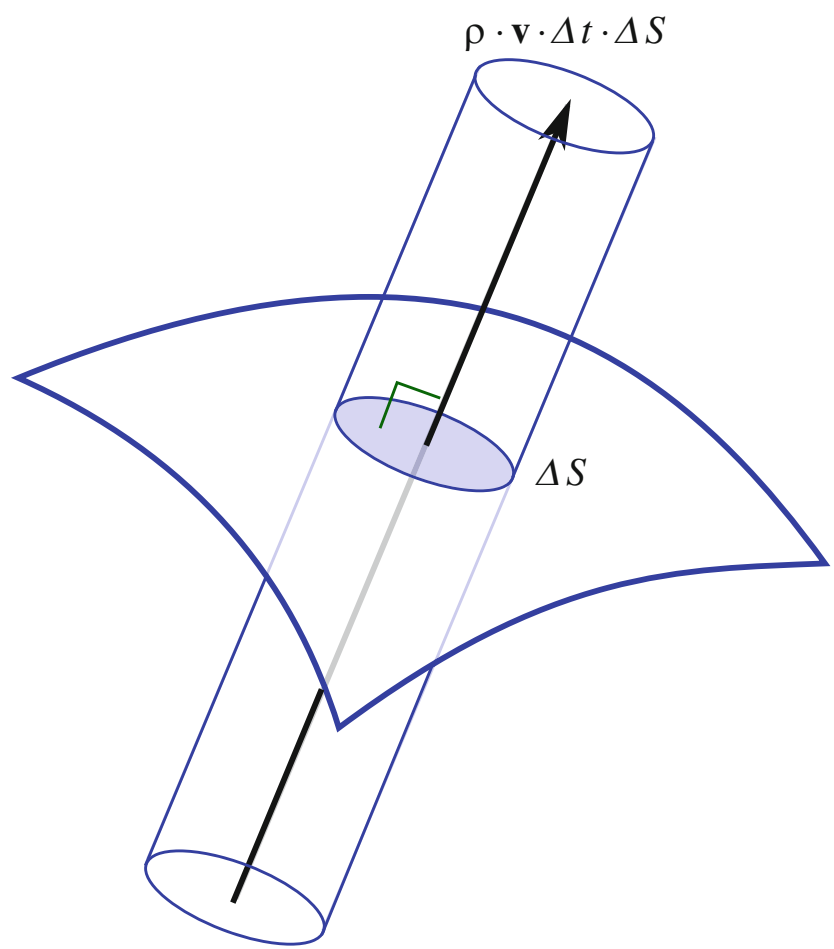



A vector $\mathbf{v}$ with base point $(x,y,z)$ that "penetrates" the surface $S$. It is helpful to imagine a particle at point $(x,y,z)$ traveling with velocity $\mathbf{v}$.




Estimating the mass of fluid flow through a small bit of surface, $\Delta S$, over a small time period $\Delta t$ assuming all particles are traveling with velocity $\mathbf{v}$ and the particles have density $\rho$.